DG INTERNS HUB
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Week :	1
Task 1:	Lab Setup,Linux and Network Foundation
Student Id:	DG/15/OCT/CYBER/118

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Linux File System Hierarchy and Permissions

1. Important Directories

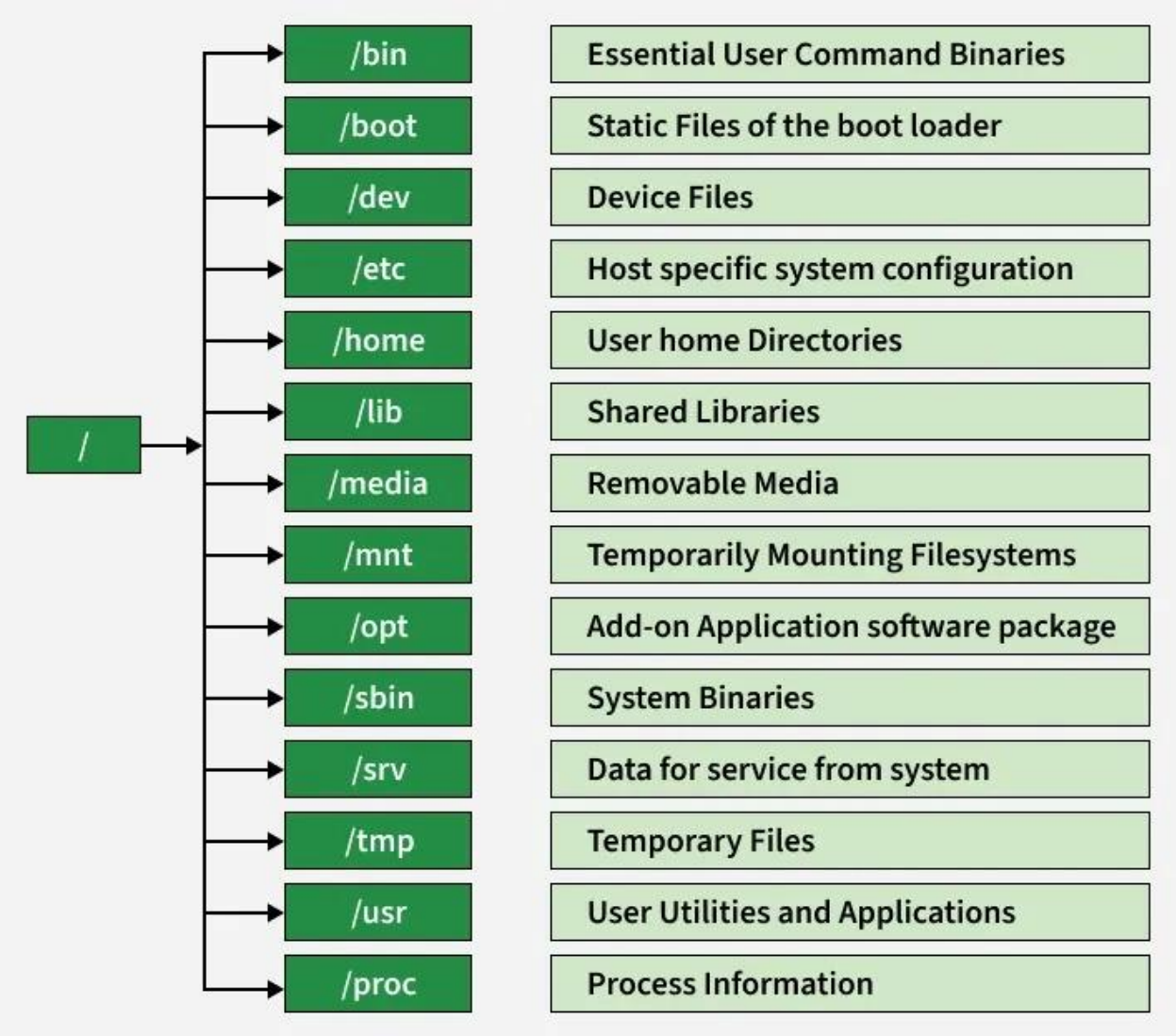
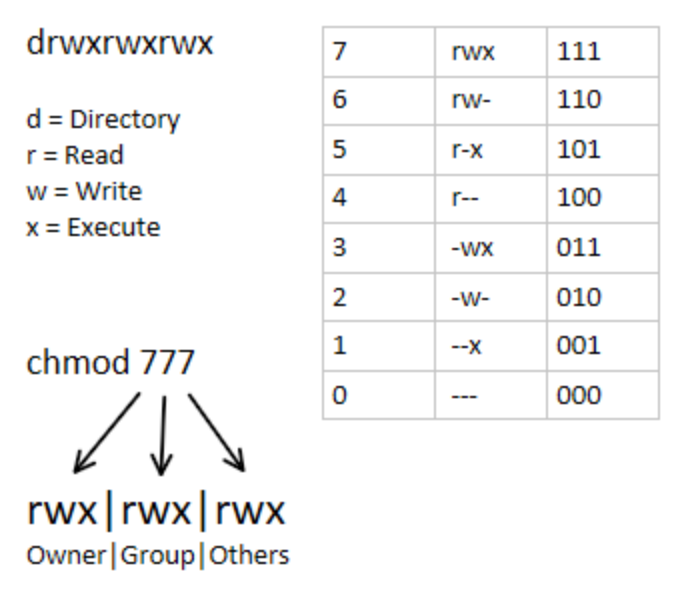


Figure 1

2. Understanding Permissions

Figure 2



7	rwx	111
6	rw-	110
5	r-x	101
4	r--	100
3	-wx	011
2	-w-	010
1	--x	001
0	---	000

3. chmod

```
┌──(kali㉿kali)-[~/Projects/dginternship]
└─$ touch report.txt

┌──(kali㉿kali)-[~/Projects/dginternship]
└─$ ls -l report.txt
-rw-rw-r-- 1 kali kali 0 Oct 22 14:40 report.txt

┌──(kali㉿kali)-[~/Projects/dginternship]
└─$ chmod 640 report.txt

┌──(kali㉿kali)-[~/Projects/dginternship]
└─$ ls -l report.txt
-rw-r----- 1 kali kali 0 Oct 22 14:40 report.txt

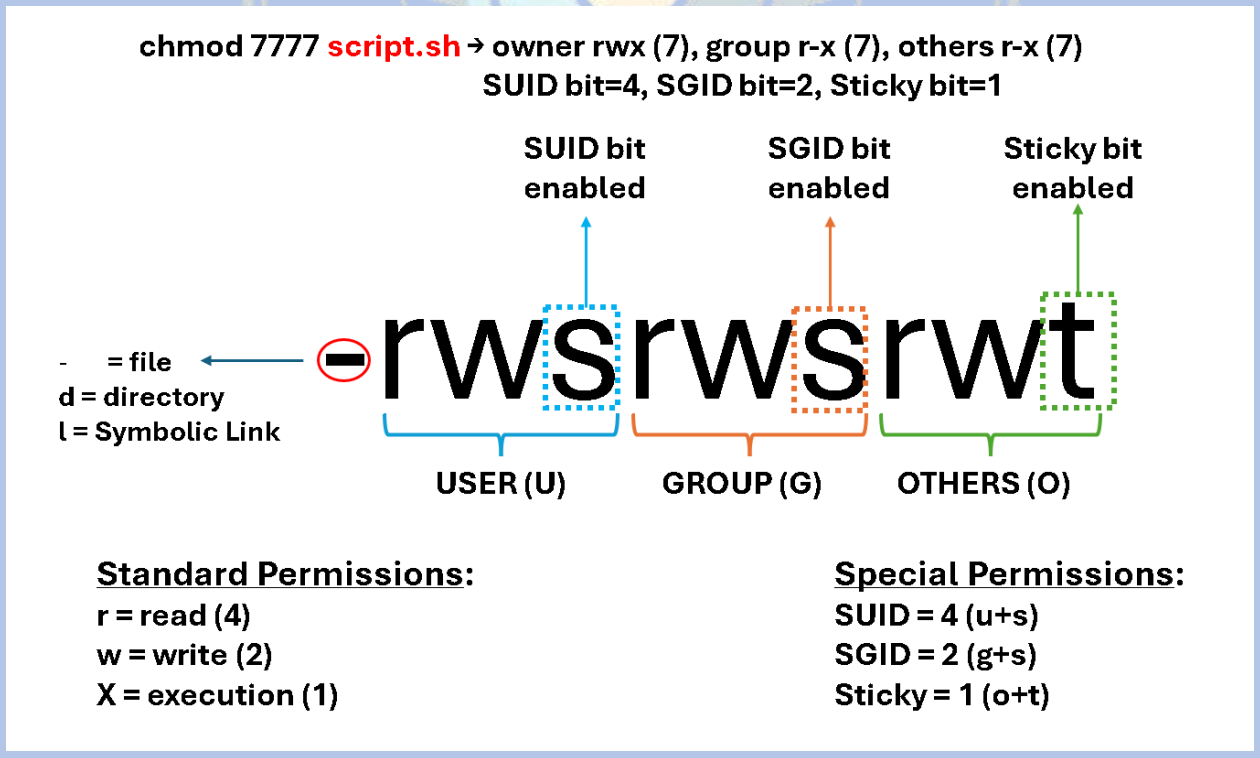
┌──(kali㉿kali)-[~/Projects/dginternship]
└─$
```

4. chown

Figure 3

```
bosko@pnap:~/Desktop$ ls -l
total 12
-rw-rw-r-- 1 root  bosko 17 Sep 26 06:27 example.txt
-rw-rw-r-- 1 bosko bosko 54 Sep 26 06:27 index.html
-r--r--r-- 1 bosko bosko  4 Sep 26 06:27 tutorial.md
bosko@pnap:~/Desktop$ sudo chown :grouptest example.txt
bosko@pnap:~/Desktop$ ls -l
total 12
-rw-rw-r-- 1 root  grouptest 17 Sep 26 06:27 example.txt
-rw-rw-r-- 1 bosko bosko     54 Sep 26 06:27 index.html
-r--r--r-- 1 bosko bosko      4 Sep 26 06:27 tutorial.md
bosko@pnap:~/Desktop$
```

5. SUID



6. Shared Directory Example

Create a group and a shared directory on an Ubuntu lab system:

Figure 4

```
┌──(kali㉿kali)-[~/Projects/dginternship]
└─$ sudo groupadd labshare
sudo mkdir -p /srv/labshare
sudo chown root:labshare /srv/labshare
sudo chmod 2770 /srv/labshare
ls -ld /srv/labshare

[sudo] password for kali:
drwxrws--- 2 root labshare 4096 Oct 22 14:49 /srv/labshare

┌──(kali㉿kali)-[~/Projects/dginternship]
└─$
```

7. Conclusion

Understanding Linux directories and permissions helps administrators protect files, review system logs and reduce unauthorized access. SOC analysts also use these concepts when investigating suspicious processes, file changes and privilege escalation.